

CLINICAL AND POPULATION SCIENCES

Intracranial and Peripheral Inflammatory Biomarkers Predict Futile Recanalization After Endovascular Treatment

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BACKGROUND: Futile recanalization (absence of functional independence despite successful endovascular treatment [EVT]) remains a limitation in the treatment of large vessel occlusion stroke. Inflammation has emerged as a key contributor to brain reperfusion injury. Leveraging EVT, we quantified a large panel of immunologic biomarkers in the intracranial and periprocedural peripheral blood samples and probed their association with futile recanalization.

METHODS: We conducted a prospective single-center cohort study of patients with anterior-circulation large vessel occlusion, successful EVT (modified Thrombolysis in Cerebral Infarction $\geq 2b$), and no reocclusion. We sampled intracranial blood distal to the thrombus before recanalization and collected peripheral blood immediately before and after EVT, and at 24 hours post-EVT. We measured 37 immune-vascular biomarkers selected a priori, using immunoassays, and tested their associations with futile recanalization (90-day modified Rankin Scale score, 3–6) using rank-based tests with false discovery rate control. Biomarkers meeting a prespecified effect-size threshold (≥ 0.5) entered multivariable logistic regression models adjusted for age, baseline National Institutes of Health Stroke Scale score and Alberta Stroke Program Early Computed Tomography Score, intravenous thrombolysis use, and time from last known well to groin puncture.

RESULTS: Among 177 enrolled patients, 139 patients were included (median age, 77 years; 41% women), and 64 (46%) had futile recanalization. Six biomarkers showed at least a medium effect size at ≥ 1 timepoints: IL (interleukin)-6 (pre-EVT, intracranial, and 24-hour post-EVT), IL-10 (pre-EVT and intracranial), IL-15 (pre-EVT), VEGFR1 (vascular endothelial growth factor receptor 1; pre-EVT and 24-hour post-EVT), VCAM-1 (vascular cell adhesion molecule-1; intracranial and post-EVT), and SAA (serum amyloid A; post-EVT). In adjusted multivariable analyses, futile recanalization remained associated with VCAM-1 intracranially (adjusted odds ratio [aOR], 1.84 [95% CI, 1.11–3.05]) and post-EVT (aOR, 2.08 [95% CI, 1.03–4.19]), IL-10 pre-EVT (aOR, 3.87 [95% CI, 1.58–9.43]) and intracranially (aOR, 4.84 [95% CI, 1.68–13.90]), IL-15 pre-EVT (aOR, 1.60 [95% CI, 1.00–2.53]), and SAA post-EVT (aOR, 7.08 [95% CI, 1.95–25.59]).

CONCLUSIONS: A core set of immune-vascular mediators measured in blood, both intracranially and peripherally, after EVT predicted futile recanalization. These findings support biomarker-guided risk stratification and motivate trials targeting microvascular inflammation.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: brain ischemia ■ endovascular procedures ■ inflammation ■ medical futility ■ reperfusion injury ■ thrombectomy

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Nonstandard Abbreviations and Acronyms

AIS	acute ischemic stroke
aOR	adjusted odds ratio
ASPECTS	Alberta Stroke Program Early Computed Tomography Score
BBB	blood-brain barrier
ECASS-II	European Cooperative Acute Stroke Study
EVT	endovascular treatment
FLT1/VEGFR1	vascular endothelial growth factor receptor 1
IL	interleukin
LVO	large vessel occlusion
MISO	Multimodal Analysis of the Origin of Intracranial Thrombi
MRI	magnetic resonance imaging
mRS	modified Rankin Scale
SAA	serum amyloid A
T24	24-hour post-endovascular treatment
VCAM-1	vascular cell adhesion molecule-1

In acute ischemic stroke (AIS) due to large vessel occlusion (LVO), endovascular treatment (EVT) achieves recanalization in over 80% of cases and improves functional outcomes.¹ However, nearly half of patients with successful recanalization fail to achieve favorable outcomes, a phenomenon widely referred to as futile recanalization.^{1–3}

Potential causes of futile recanalization include early reocclusion and reperfusion injury, the latter encompassing parenchymal hemorrhage, edema, and the no-reflow phenomenon, defined as the failure of microvascular reperfusion despite complete arterial recanalization.^{4–6} These components of reperfusion injury have been associated, in preclinical studies, with potentially targetable inflammatory processes.^{4,7–9} Hence, inflammation contributes to blood-brain barrier (BBB) damage and increases the risk of hemorrhagic transformation by elevating reactive oxygen species and matrix metalloproteinases, and neutrophil activation.⁷ Inflammation has also been implicated in vasogenic edema formation through cytokine-mediated BBB permeability and astrocyte swelling or activation.⁹ Finally, inflammation plays a central role in the development of no-reflow, notably through neutrophil-mediated capillary obstruction, endothelial swelling, vasogenic edema-related capillary compression, and pericyte constriction.^{8,10}

Emerging clinical studies have examined the role of inflammation in futile recanalization, with most focusing on immune cell measures such as neutrophil, monocyte, or lymphocyte counts, the neutrophil-to-lymphocyte ratio,

or C-reactive protein.^{11–18} Processes involved in reperfusion injury can initiate during the ischemic phase itself, some almost at stroke onset,⁴ supporting the rationale for exploring inflammatory pathways at early timepoints, including during EVT. Emerging studies that collected arterial blood adjacent to the intracranial thrombus during EVT reported cytokines and chemokines significantly associated with AIS infarct and edema volumes. Others found that VCAM-1 (vascular cell adhesion molecule-1) was correlated with poor early neurological improvement.^{19,20} In our previous work, using intracranial blood samples obtained during EVT before recanalization, we found that higher levels of IL (interleukin)-6 and chemokine ligand 17 distal to the clot were moderately but significantly associated with poorer outcomes.²¹ These findings supported a potential role for local inflammation in clinical outcomes although available studies did not focus on futile recanalization or systematically excluded clinical confounding factors. We also performed an exploratory study on the temporal dynamics of immunologic trajectories specifically in the peripheral blood,²² from pre- to post-EVT, which supports the hypothesis that processes involved in reperfusion injury may initiate during the ischemic phase and rapidly evolve after EVT.⁴

Building on this evidence, we leveraged a large cohort with intracranial and serial peripheral blood sampling to profile an extensive panel of cytokines, chemokines, and vascular factors, thereby pinpointing inflammation's role in futile recanalization and its contribution to reperfusion injury. We focused our analysis on patients with successful and sustained recanalization, excluding those with documented early reocclusion. In a sensitivity analysis, we further excluded patients who developed parenchymal hematoma to specifically explore the relationship between inflammatory biomarkers and other components of reperfusion injury.

METHODS

Data Availability

The data supporting the study findings are available from the corresponding author upon reasonable request.

Study Design and Population

This was a single-center observational study including patients from the prospective MISO (Multimodal Analysis of the Origin of Intracranial Thrombi) cohort (RECHMPL1880236, ID-RCB 2018-A02651-54).^{21,22} The primary objective of the MISO cohort was to identify biological biomarkers associated with radiological and clinical outcomes by collecting both peripheral and intracranial blood samples during EVT.²¹ The inclusion criteria of MISO were (1) age ≥ 18 years, (2) AIS due to intracranial LVO treated by EVT at Montpellier University Hospital, (3) absence of known active infection or ongoing antibiotic treatment at the time of the EVT, and (4) acquisition of blood samples during EVT.²¹ Starting in August 2023, only patients

with successful recanalization (as assessed by the neuroradiologist in charge) of an anterior LVO were included in MISO (the evolution of MISO inclusion criteria over time is summarized in Figure S1).

Patients from the MISO cohort were eligible in this study if they fulfilled the following criteria: (1) prestroke modified Rankin Scale (mRS) score ≤ 2 ; (2) AIS due to intracranial LVO of the anterior circulation, defined as an occlusion of the first segment of the middle cerebral artery or second segment of the middle cerebral artery, or the intracranial internal carotid artery; (3) successful recanalization as assessed by an independent neuroradiologist (see below); (4) absence of intracranial reocclusion on a follow-up imaging during hospitalization (see below); and (5) available 3-month mRS score. Exclusion criteria were (1) active malignancies and (2) inflammatory disease under chronic anti-inflammatory treatment.

Patient Management

All patients had confirmed AIS due to LVO on baseline brain imaging (computed tomography with angiography or magnetic resonance imaging [MRI] with angiography). Intravenous thrombolysis was administered according to current guidelines.²³ EVT was performed using stent retrievers and aspiration techniques according to standard neurointerventional protocols, as previously described.²¹ A 24-hour follow-up brain imaging, including arterial assessment, was performed for most patients. This follow-up imaging modality at 24 hours varied across the study period (computed tomography in the first phase and MRI in the second). Starting in August 2023, an early postthrombectomy MRI with time-of-flight was additionally performed, accounting for 90 patients in the present study.

Outcome Assessment and Definition

Recanalization status was independently assessed by an experienced neuroradiologist (C.D.), blinded to all clinical and biological data, using final digital subtraction angiography runs. Only patients with successful recanalization, defined as modified Thrombolysis in Cerebral Infarction $\geq 2b$, were included.

Post-EVT intracranial reocclusion was defined as a reocclusion at the site of the previous recanalized occlusion or a new intracranial proximal occlusion and was assessed by an experienced stroke neurologist (A.t.S.) on each follow-up arterial imaging performed during hospitalization, including 24-hour imaging, and early postthrombectomy imaging when available. As stated before, patients with reocclusion were excluded from the study.

Futile recanalization was defined as a 90-day mRS score of 3 to 6, as in previous studies.^{2,24} The mRS score was determined by a stroke neurologist or neuroradiologist during in-person or phone follow-up.

Hemorrhagic transformation was assessed on 24-hour imaging and early postthrombectomy (when performed), using ECASS-II (European Cooperative Acute Stroke Study) classification,²⁵ jointly by a stroke neurologist (A.t.S.) and an interventional neuroradiologist (C.D.), with consensus in case of discrepancies.

Blood Sampling and Processing

Peripheral blood samples were collected at 3 timepoints: before EVT during femoral artery puncture (pre-EVT), upon femoral

catheter withdrawal after recanalization (post-EVT), and at 24 hours post-recanalization (24-hour post-EVT [T24]). Post-EVT and T24 samples were not systematically performed in the earlier phase of the cohort, and T24 samples were not collected in patients placed under palliative care within the first 24 hours or in drip-and-ship patients who were repatriated early to their primary stroke center.

An intracranial blood sample was collected in patients treated with a stent retriever via the microcatheter positioned distal to the occlusion site and before recanalization, allowing targeted analysis of the local ischemic environment, as previously described.²¹ In some patients, intracranial blood collection could not be performed, either due to absent blood reflux or because aspiration alone was used, precluding microcatheter access beyond the clot.

A panel of 37 inflammatory biomarkers, including cytokines, chemokines, and adhesion molecules, selected a priori based on the preclinical literature,^{7–9,24,26–30} was quantified using multiplex immunoassays (V-Plex Neuroinflammation Panel 1 Human Kit, Cat K15210D, MSD). Values under the limit of detection were considered as zero values in analyses. Details regarding sample handling, timing, and biomarkers are provided in the [Supplemental Methods](#).

Other Collected Data

The following additional data were collected: (1) clinical data: age, sex, prestroke mRS score, vascular risk factors, time of symptom onset or last known well, baseline and 24-hour National Institutes of Health Stroke Scale score, and stroke cause based on the Trial of ORG 10172 in Acute Stroke Treatment classification; (2) radiological data: occlusion site on digital subtraction angiography, baseline, and 24-hour Alberta Stroke Program Early Computed Tomography Score (ASPECTS); and (3) treatment data: intravenous thrombolysis realization and time, EVT puncture time, and number of device passes.

Standard Protocol Approvals, Registrations, and Patient Consents

The MISO cohort was approved by the Montpellier University Hospital (RECHMPL1880236, CPP 2019.10.04_19.07.30.36737).²¹ Informed consent was obtained from all participants through a consent form. This study was approved by the ethics committee and performed according to the standards of the Declaration of Helsinki. Reporting was performed according to the STROBE statement (Strengthening the Reporting of Observational Studies in Epidemiology; [Supplemental Material](#)).

Statistical Methods

Continuous variables were reported as medians with interquartile ranges, and categorical variables were reported as counts and percentages. Baseline and EVT characteristics were described and compared according to the presence or absence of futile recanalization. Quantitative variables were compared using the Student *t* test for normally distributed data or the Mann-Whitney *U* test otherwise, and categorical variables using the χ^2 test or the Fisher exact test when expected cell counts were < 5 . We also assessed a potential selection bias in analyses

due to missing inflammatory biomarker measurements at intracranial, post-EVT, and T24 timepoints by reporting baseline and EVT characteristics in patients with and without missing values (Tables S1 through S3).

We first investigated the association between futile recanalization and inflammatory biomarkers assessed at each timepoint (pre-EVT, intracranial, post-EVT, and T24) by comparing the biomarkers' distributions (most highly skewed) according to the presence or absence of futile recanalization using the Mann-Whitney *U* tests. To account for multiple testing, *P* values were adjusted using the Benjamini-Hochberg procedure at each timepoint to control the false discovery rate. We also evaluated the magnitude of between-group differences by calculating standardized mean differences (effect sizes) on rank-transformed data, which were reported in forest plots. Inflammatory biomarkers related to futile recanalization with an effect size ≥ 0.5 (interpreted as medium effect size³¹ and considered as biomarkers of interest) were further investigated in logistic regression models with and without adjustment for predefined confounders known to be associated with futile recanalization in the literature^{2,16} (namely, age, baseline National Institutes of Health Stroke Scale score and ASPECTS, intravenous thrombolysis use, and time from last known well to groin puncture). To reduce the skewness of distributions, all biomarkers were log-transformed, and the log-linearity assumption was assessed using restricted cubic splines. Biomarkers with a non-log-linear association (with a significant spline term at $P < 0.10$) were further analyzed as binary variables using the cutoff point that maximizes the Youden index from receiver operating characteristic curve analysis. Unadjusted and adjusted odds ratios (aORs) were estimated from logistic regression models as an effect size.

In a prespecified sensitivity analysis, we excluded patients with parenchymal hematoma to evaluate whether the associations between biomarkers and futile recanalization were driven primarily by overt hematoma-related clinical deterioration. This analysis was intended as a complementary exploration of biomarker associations with nonhemorrhagic components of reperfusion injury although we acknowledge that parenchymal hematoma is itself a key manifestation of reperfusion injury.

In an exploratory analysis addressing the relationship between dynamic changes of the biological biomarkers of interest and futile recanalization, we calculated the relative change of post-EVT values and T24 values compared with pre-EVT values. We then compared these changes according to the presence or absence of futile recanalization using the Mann-Whitney *U* tests and calculated the standardized mean differences on rank-transformed data as effect sizes.

Finally, in a last exploratory analysis addressing the relationship between 24-hour inflammatory biomarker levels and ischemic lesion extent at the same timepoint, we assessed correlations between T24 biomarkers' concentrations and 24-hour ASPECTS using the Spearman rank correlation coefficients (95% CIs were estimated using the Fisher *z* transformation). Statistical testing was done at the 2-tailed α level of 0.05. All analyses were based on available data without imputing missing values. Data were analyzed using the SAS software package, release 9.4 (SAS Institute, Cary, NC).

RESULTS

Patient Characteristics

Between March 1, 2021, and November 15, 2024, 177 patients were enrolled in the MISO cohort. Among them, 38 were excluded from the present analysis due to intracranial reocclusion during hospitalization ($n=12$), modified Thrombolysis in Cerebral Infarction recanalization $\leq 2b$ (as assessed post hoc by an independent neuroradiologist; $n=8$), prestroke mRS score >2 ($n=7$), active malignancies or inflammatory diseases under anti-inflammatory treatment ($n=6$), posterior circulation LVO ($n=3$), and missing 90-day mRS score ($n=2$; see flowchart in Figure 1).

A total of 139 patients were included, of whom 64 (46%) experienced futile recanalization (Table 1). Patients with futile recanalization were older (median,

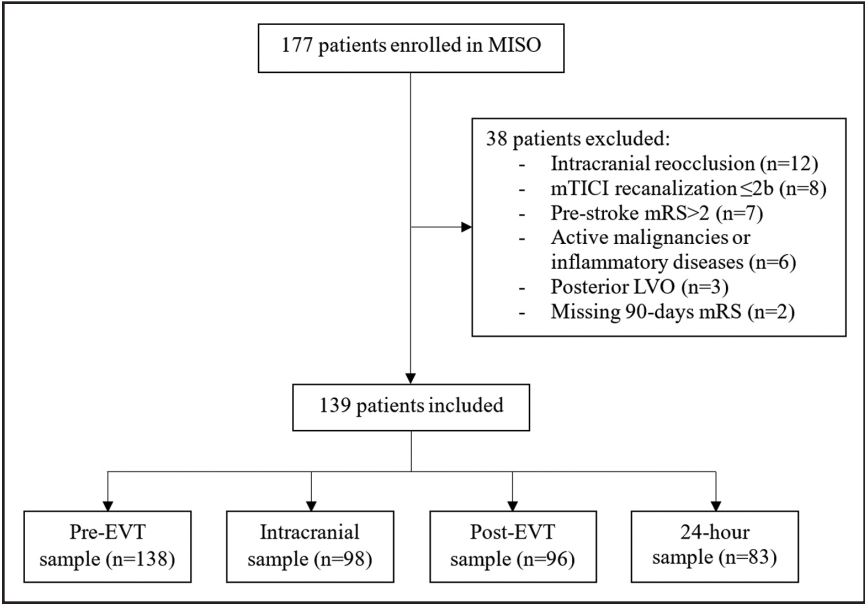


Figure 1. Flowchart of patient inclusion and blood sampling timepoints.

Regarding sample timepoints, from March 2023, intracranial samples were no longer mandatory and were collected on a best-effort basis (Figure S1). Some post-endovascular treatment (EVT) and 24-hour samples were not collected during the early phase of the study, which accounts for the smaller numbers at these timepoints. LVO indicates large vessel occlusion; MISO, Multimodal Analysis of the Origin of Intracranial Thrombi; mRS, modified Rankin Scale; and mTICI, modified Thrombolysis in Cerebral Infarction.

Table 1. Characteristics of Study Sample, Overall, and According to Futile Recanalization Status

	Overall	Futile recanalization		
Characteristics	(N=139)	No (n=75)	Yes (n=64)	P value
Baseline demographics and medical history				
Age, y	77 (70–83)	75 (66–81)	81 (71–85)	0.017
Men	82 (59)	47 (63)	35 (55)	0.34
Hypertension	86 (62)	41 (55)	45 (70)	0.058
Diabetes	19 (14)	9 (12)	10 (16)	0.54
Hypercholesterolemia	48 (35)	21 (28)	27 (42)	0.080
Current smoking	17 (12)	12 (16)	5 (8)	0.14
History of stroke or TIA	20 (14)	7 (9)	13 (20)	0.066
Coronary artery disease	15 (11)	7 (9)	8 (13)	0.55
Current stroke event				
Admission glucose, mmol/L	6.6 (5.8–7.7)	6.2 (5.7–7.4)	6.8 (6.1–8.4)	0.075
Admission NIHSS score	17 (10–22)	13 (8–21)	19 (14–22)	<0.001
Admission ASPECTS	8 (7–9)	8 (7–10)	7 (5–9)	0.001
Intracranial occlusion site				0.047
M1-MCA	84 (60)	47 (63)	37 (58)	
M2-MCA	21 (15)	15 (20)	6 (9)	
Terminal ICA	34 (25)	13 (17)	21 (33)	
Tandem occlusion	18 (13)	8 (11)	10 (16)	0.39
Suspected stroke cause				0.78
Cardioembolic	82 (59)	44 (59)	38 (59)	
Large artery atherosclerosis	18 (13)	11 (15)	7 (11)	
Other/unknown	39 (28)	20 (27)	19 (30)	
Intravenous thrombolysis	61 (44)	43 (57)	18 (28)	<0.001
EVT characteristics				
Last known well to groin puncture time, min	341 (205–729)	301 (190–478)	404 (243–848)	0.012
Groin puncture to recanalization, min	25 (18–44)	25 (17–44)	30 (19–44)	0.26
mTICI				
2b	48 (35)	29 (39)	19 (30)	0.26
2c-3	91 (65)	46 (61)	45 (70)	
24-h hemorrhages	n=137		n=62	
Any intracerebral hemorrhage	78 (57)	38 (51)	40 (67)	0.10
SAH	18 (13)	11 (15)	7 (11)	0.56
HI1-HI2	41 (30)	22 (29)	19 (31)	0.87
PH1-PH2	19 (14)	5 (7)	14 (23)	0.007

Values expressed as number (%) or median (25th–75th percentiles; n=number of available patients). ASPECTS indicates Alberta Stroke Program Early Computed Tomography Score; EVT, endovascular treatment; HI, hemorrhagic infarction; ICA, internal carotid artery; M1, first segment of the middle cerebral artery; M2, second segment of the middle cerebral artery; MCA, middle cerebral artery; mTICI, modified Thrombolysis in Cerebral Infarction; NIHSS, National Institutes of Health Stroke Scale; PH, parenchymal hematoma; SAH, subarachnoid hemorrhage; and TIA, transient ischemic attack.

81 versus 75 years), had higher admission National Institutes of Health Stroke Scale scores (median, 19 versus 13), and lower ASPECTS scores (median, 7 versus 8). They were also less likely to have received intravenous thrombolysis (28% versus 57%). They had longer delays from last known well to groin puncture (median, 404 versus 301 minutes). In addition, parenchymal hematomas were more frequently observed in this group (22% versus 7%). Other baseline characteristics, such as sex, vascular risk factors, stroke cause,

and groin puncture to recanalization time, were well balanced between groups.

Immuno-Vascular Biomarkers and Futile Recanalization

We first cross-compared the levels of 37 inflammatory biomarkers between patients with and without futile recanalization in intracranial and peripheral samples: pre-EVT (n=138), intracranial (n=98), post-EVT (n=96),

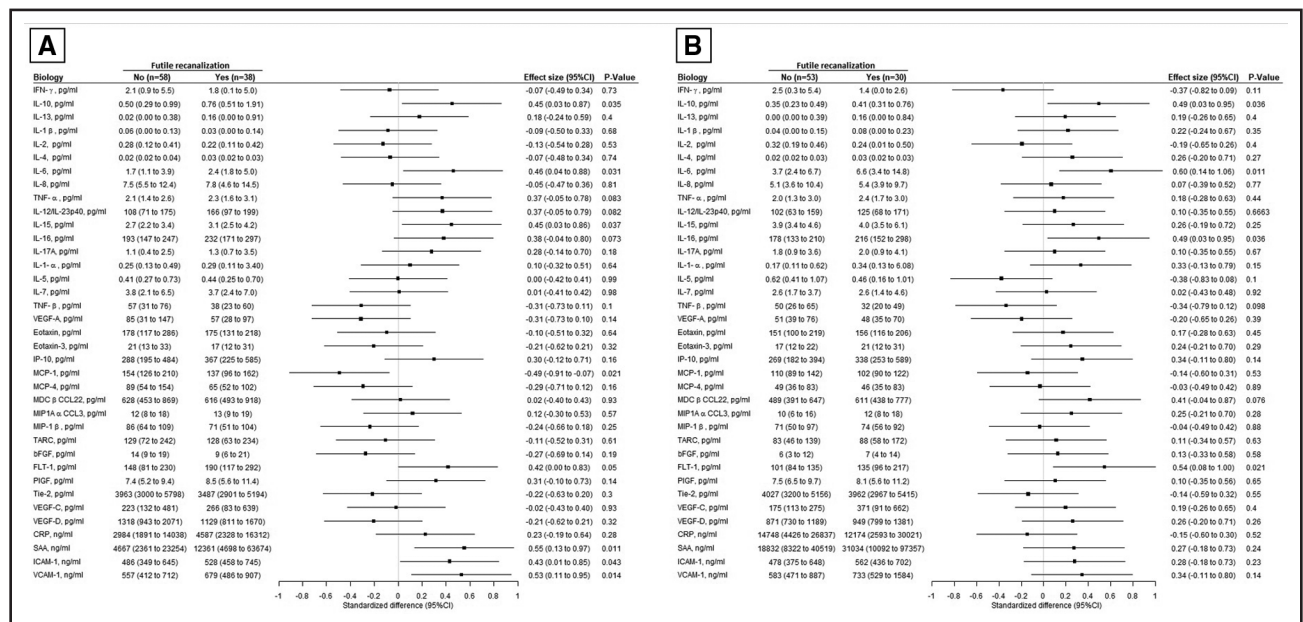
and T24 (n=83). The results are presented in Figures 2 and 3. Numerous biomarkers were associated with futile recanalization, several of which remained significantly associated after applying false discovery rate correction: IL-6, IL-10, IL-15, IL-16, and FLT1/VEGFR1 (vascular endothelial growth factor receptor 1) at the pre-EVT timepoint, and IL-6, IL-10, and VCAM-1 for the intracranial sample (no association was found for post-EVT and 24-hour timepoints).

Regarding between-group differences in biomarker distributions according to futile recanalization, 6 biomarkers were associated with at least a medium effect size (standardized mean difference ≥ 0.5) at ≥ 1 timepoints: IL-6, IL-10, IL-15, and FLT1/VEGFR1 at the pre-EVT timepoint; IL-10, IL-6, and VCAM-1 intracranially; SAA (serum amyloid A) and VCAM-1 at the post-EVT timepoint; and IL-6 and FLT1/VEGFR1 at the 24-hour timepoint. We further investigated the associations between these immune-vascular factors and futile recanalization in logistic regression models adjusted for key clinical confounders (age, National Institutes of Health Stroke Scale score, ASPECTS, last known well to groin puncture time, and intravenous thrombolysis use; Table 2).

In the pre-EVT peripheral blood, higher levels of IL-10 were independently associated with futile recanalization (aOR, 3.87 [95% CI, 1.58–9.43] for IL-10 >0.49 pg/mL). Regarding intracranial samples, IL-10 (aOR, 4.84 [95% CI, 1.68–13.90] for IL-10 >0.46 pg/mL) and VCAM-1 (aOR, 1.84 [95% CI, 1.11–3.05]) remained significant. Post-EVT elevated levels of SAA (aOR, 7.08 [95% CI, 1.95–25.59] for SAA >8600 pg/mL) and VCAM-1 (aOR, 2.08 [95% CI, 1.03–4.19]) were also independently associated with futile outcomes. When sampled 24 hours after EVT, none of the studied biomarkers remained significantly associated with futile recanalization after adjustment.

Sensitivity Analysis Excluding Patients With PH

We performed a sensitivity analysis including 120 patients after excluding the 19 patients with parenchymal hematoma. The results of the association between immune-vascular factors and futile recanalization on univariate analysis are presented in Figures S2 and S3. Numerous biomarkers were associated with futile recanalization, but none remained significantly associated after applying false discovery rate correction. Regarding the magnitude



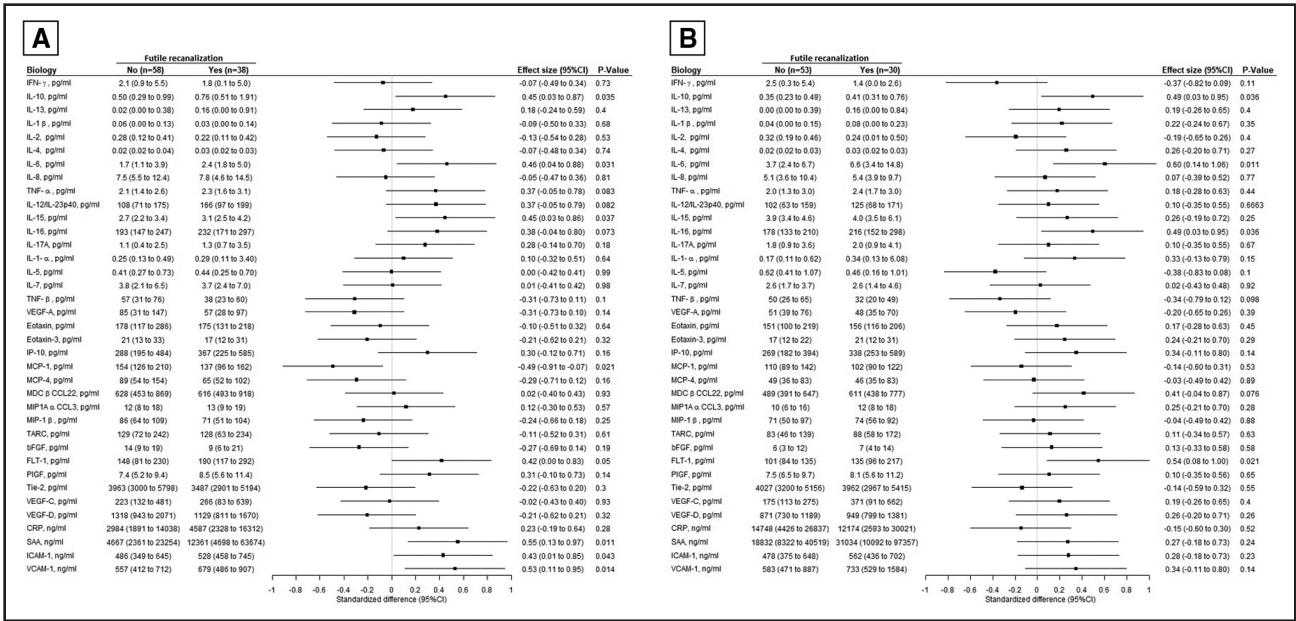


Figure 3. Comparison of distributions of immune-vascular biomarkers assessed at post-endovascular treatment and 24-hour post-endovascular treatment (T24) timepoints according to futile recanalization status.

A, Post-endovascular treatment; **B** T24. Standardized mean differences calculated on rank-transformed data were reported as effect sizes; effect sizes ≥ 0.5 were interpreted as medium effect sizes and considered as potential biomarkers of interest. *P* values were calculated using the Mann-Whitney *U* tests. bFGF indicates basic fibroblast growth factor; CRP, C-reactive protein; FLT1, fms-related receptor tyrosine kinase 1 (also known as VEGFR1 [vascular endothelial growth factor receptor 1]); ICAM-1, intercellular adhesion molecule-1; IFN- γ , interferon gamma; IL, interleukin; IP-10, interferon gamma-induced protein 10 (also known as CXCL10 [C-X-C motif chemokine ligand 10]); MCP-1, monocyte chemoattractant protein-1 (also known as CCL2 [chemokine CC-motif ligand 2]); MCP-4, monocyte chemoattractant protein-4 (also known as CCL13 [chemokine CC-motif ligand 13]); MDC, macrophage-derived chemokine (also known as CCL22 [chemokine CC-motif ligand 22]); MIP1- α , macrophage inflammatory protein alpha (also known as CCL3 [chemokine CC-motif ligand 3]); MIP-1 β , macrophage inflammatory protein beta (also known as CCL4 [chemokine CC-motif ligand 4]); PlGF, placenta growth factor; SAA, serum amyloid A; TARC, thymus and activation-regulated chemokine (also known as CCL17 [chemokine CC-motif ligand 17]); Tie-2, tyrosine kinase with immunoglobulin and epidermal growth factor homology domains; TNF- α , tumor necrosis factor alpha; VCAM-1, vascular cell adhesion molecule-1; and VEGF, vascular endothelial growth factor. **P* < 0.05 after applying the Benjamini-Hochberg procedure to control the false discovery rate (FDR).

of between-group differences in biomarker distributions according to futile recanalization, 4 biomarkers were associated with at least a medium effect size (standardized mean difference ≥ 0.5): IL-6 in the pre-EVT peripheral sample. In the adjusted models, intracranial IL-10 (aOR, 7.73 [2.03–29.49] for IL-10 > 0.35 pg/mL), SAA (aOR, 4.62 [1.37–15.62] for SAA > 3100 pg/mL), and VCAM-1 (aOR, 1.97 [1.22–3.46]) remained independently associated with futile recanalization. In contrast, IL-6 was no longer significant after adjustment (Table 3).

Dynamic Changes of Selected Mediators

In exploratory analyses, percent relative changes from pre-EVT to post-EVT and from pre-EVT to 24 hours did not differ significantly between patients with and without futile recanalization (Figure S4).

Correlation Between 24-Hour Biomarkers and 24-Hour ASPECTS

In exploratory analyses including the 83 patients with both 24-hour biomarker analyses and 24-hour

follow-up imaging, no significant correlations were observed between T24 biomarker levels and 24-hour ASPECTS after applying false discovery rate correction (Figure S5).

DISCUSSION

In this prospective monocentric study, we identified several blood-based immune-vascular factors independently associated with futile recanalization after adjustment for key clinical confounders. Among the 37 tested biomarkers, VCAM-1 and IL-10 emerged as the most consistent predictors of futile recanalization. Importantly, both biomarkers were significantly associated with poor outcomes when measured in intracranial samples before recanalization, within the ischemic territory, and either immediately before (IL-10) or immediately after (VCAM-1) thrombectomy for peripheral blood samples. This combination of sampling site and timing is consistent with the notion that early, localized immune responses are active before recanalization and may directly contribute to reperfusion injury, including hemorrhagic transformation.⁴ Importantly, these associations remained

Table 2. Unadjusted and Adjusted ORs for the Selected Biomarkers*

Biomarkers	Unadjusted OR† (95% CI)	P value	Adjusted OR† (95% CI)	P value
Pre-EVT biomarker values				
IL-10 >0.49 pg/mL‡	4.17 (2.00–8.66)	<0.001	3.87 (1.58–9.43)	0.003
IL-6	1.84 (1.23–2.75)	0.003	1.21 (0.75–1.93)	0.43
IL-15	1.83 (1.26–2.77)	0.002	1.60 (1.00–2.53)	0.047
FLT1/VEGFR1	2.12 (1.32–3.28)	0.002	1.55 (0.92–2.60)	0.098
Intracranial biomarker values				
IL-10 >0.46 pg/mL‡	4.31 (1.84–10.07)	<0.001	4.84 (1.68–13.90)	0.003
IL-6	1.88 (1.19–2.97)	0.007	1.54 (0.87–2.70)	0.13
VCAM-1	1.92 (1.20–3.05)	0.006	1.84 (1.11–3.05)	0.018
Post-EVT biomarker values				
SAA >8600 pg/mL‡	3.52 (1.49–8.30)	0.004	7.08 (1.95–25.59)	0.003
VCAM-1	2.35 (1.29–4.31)	0.005	2.08 (1.03–4.19)	0.041
T24 biomarker values				
IL-6	1.92 (1.13–3.24)	0.015	1.49 (0.79–2.78)	0.22
FLT1/VEGFR1 >110 pg/mL‡	3.59 (1.35–9.51)	0.010	2.17 (0.65–7.17)	0.20

ASPECTS indicates Alberta Stroke Program Early Computed Tomography Score; EVT, endovascular treatment; FLT1/VEGFR1, vascular endothelial growth factor receptor 1; IL, interleukin; NIHSS, National Institutes of Health Stroke Scale; OR, odds ratio; ROC, receiver operating characteristic; SAA, serum amyloid A; T24, 24-hour post-endovascular treatment; and VCAM-1, vascular cell adhesion molecule-1.

*Biomarkers selected based on a standardized mean difference ≥ 0.5 in the biomarker distribution between patients with and without futile recanalization.

†OR indicates the odds ratio of futile recanalization per 1-SD increase in log-transformed biomarker values (except for IL-10, SAA, and FLT1/VEGFR1 at 24-hour post-EVT values where a non-log-linear association was observed), estimated from logistic regression models before and adjusted for prespecified confounders (age, intravenous thrombolysis, last known well to groin puncture time, pretreatment NIHSS score, and ASPECTS scores).

‡Cutoff point that maximizes the Youden index from ROC curve analysis.

significant in a sensitivity analysis excluding patients with parenchymal hematoma, supporting their link to nonhemorrhagic components of reperfusion injury such as no-reflow or cerebral edema. SAA (post-EVT) and IL-15 (pre-EVT) were also associated with futile recanalization after adjustment, with intracranially measured SAA being significant in sensitivity analyses, supporting

its role in nonhemorrhagic components of reperfusion injury. This sensitivity analysis should, nonetheless, be interpreted cautiously. Indeed, some mediators may promote futility by inducing no-reflow, as well as hemorrhagic transformation. Therefore, excluding patients with parenchymal hematoma may remove part of the biological signal of interest and paradoxically reduce our ability

Table 3. Unadjusted and Adjusted ORs for the Selected Biomarkers* in Sensitivity Analyses Excluding Patients With Parenchymal Hematoma

Biomarkers	Unadjusted OR† (95% CI)	P value	Adjusted OR† (95% CI)	P value
Pre-EVT biomarker values				
IL-6	1.70 (1.13–2.55)	0.010	0.99 (0.59–1.65)	0.97
Intracranial biomarker values				
IL-10 >0.35 pg/mL‡	5.90 (1.96–17.99)	0.002	7.73 (2.03–29.49)	0.003
IL-6	1.75 (1.08–2.83)	0.022	1.25 (0.68–2.14)	0.47
SAA >3100 pg/mL‡	4.56 (1.76–11.81)	0.002	4.62 (1.37–15.62)	0.014
VCAM-1	2.09 (1.24–3.51)	0.006	1.97 (1.22–3.46)	0.018

ASPECTS indicates Alberta Stroke Program Early Computed Tomography Score; EVT, endovascular treatment; IL, interleukin; NIHSS, National Institutes of Health Stroke Scale; OR, odds ratio; ROC, receiver operating characteristic; SAA, serum amyloid A; and VCAM-1, vascular cell adhesion molecule-1.

*Biomarkers selected based on a standardized difference ≥ 0.5 in the biomarker distribution between patients with and without futile recanalization after exclusion of patients with parenchymal hematoma.

†OR indicates the odds ratio of futile recanalization per 1-SD increase in log-transformed biomarker values (except for IL-10 and SAA where a non-log-linear association was observed), estimated from logistic regression models before and adjusted for prespecified confounders (age, intravenous thrombolysis, last known well to groin puncture time, pretreatment NIHSS score, and ASPECTS scores).

‡Cutoff point that maximizes the Youden index from ROC curve analysis.

to detect meaningful mediator releases that underlie futility driven by microvascular dysfunction. Nevertheless, our study offers a unique, biological perspective on the ischemic microenvironment, complementing the other radiological predictors of futile recanalization provided by routinely available brain imaging. Thus, while MRI can detect hemorrhagic transformation, vasogenic edema, perfusion deficits, and BBB disruption at the tissue level,^{5,6} it lacks the spatial resolution to capture capillary-level dysfunction.

Regarding VCAM-1, whose elevated levels in intracranial and post-EVT samples were associated with poor outcomes, preclinical evidence strongly supports its role in reperfusion injury. VCAM-1 is primarily synthesized by activated endothelial cells in response to hypoxia and proinflammatory cytokines. It mediates leukocyte adhesion and transmigration,³² 2 mechanisms known to occur during, and probably contribute to, no-reflow.²⁶ In vitro, VCAM-1 has been shown to increase the permeability of brain endothelial cells, leading to BBB damage,³³ a role further supported by in vivo findings of increased vascular leakage in models of chronic cerebral hypoperfusion.²⁷ Finally, VCAM-1 has also been explored as a molecular imaging target, with VCAM-1–targeted MRI contrast agents successfully detecting endothelial activation and neuroinflammation in experimental stroke models.³⁴

In human stroke, several recent studies have reported that elevated VCAM-1 levels were associated with poor prognosis, including functional outcome or 3-month/1-year mortality.^{27,35,36} However, unlike our study, these assessments were performed on peripheral blood within the first 24 hours after stroke onset, and VCAM-1 levels increase during the first 48 hours after EVT.²² More recently, Maglinger et al²⁰ analyzed postclot and systemic blood samples from 40 patients undergoing thrombectomy and found that elevated intracranial VCAM-1 correlated with larger infarct and edema volumes, as well as poor early neurological improvement. Our study extends these findings by demonstrating a robust association between intracranial and post-EVT VCAM-1 levels and futile recanalization.

VCAM-1 also represents a potential therapeutic target. In mouse models of ischemic stroke, blockade of VCAM-1 or its receptor attenuated neuroinflammatory responses, reduced lymphocyte infiltration, and restored BBB integrity.^{37,38} Rather than blocking VCAM-1, an alternative and innovative approach could involve leveraging its expression on activated brain endothelium as a gateway for targeted drug delivery.³⁹ A recent preclinical study demonstrated that lipid nanoparticles functionalized with a VCAM-1–binding peptide, and containing either IL-10 mRNA or dexamethasone, selectively accumulated in the ischemic brain at concentrations 10- to 100-fold higher than untargeted nanoparticles. This targeting enabled delivery to key cellular players in poststroke inflammation, including activated endothelial cells and infiltrating

leukocytes, and, importantly, led to substantially reduced infarct volume.⁴⁰ Future research should further explore VCAM-1–targeted strategies.

IL-10, described as an anti-inflammatory cytokine, showed a somewhat unexpected positive association with futile recanalization in our study, specifically in pre-EVT and intracranial samples. These findings seem, at first glance, to contrast with the large body of preclinical literature consistently showing a protective role of IL-10 in ischemic stroke.^{28,29} Studies have reported that IL-10 administration in rodent models reduces brain injury following stroke, while IL-10–deficient rodents develop larger infarcts, exacerbated neuroinflammation, and worse long-term outcomes.^{40–44} In humans, results have, however, been conflicting.^{45,46} In a cohort of patients with mild infarcts treated mostly without recanalization therapies, reduced serum IL-10 levels within 24 hours after admission were independently associated with poor functional outcomes.⁴⁶ Conversely, in another study, higher IL-10 levels measured at 48 hours independently predicted adverse 3-month outcomes,⁴⁵ and another showed that higher IL-10 levels sampled within 24 hours after stroke were associated with 1-year mortality.³⁶ These discrepancies may stem from differences in sampling compartments, patient profiles, and timing of IL-10 assessment. In line with this, our previous analysis revealed dynamic changes in IL-10 levels across the peri-EVT period, with early elevations contrasting with later decreases.²² In our study, the association between elevated IL-10 levels and futile recanalization was observed in patients with severe LVO-related strokes at the prerecanalization phase. Importantly, this observation should not be interpreted as evidence that IL-10 is deleterious. Rather, in the hyperacute setting, early IL-10 elevations may reflect the intensity of the inflammatory stimulus and a compensatory counterregulatory response that could be insufficient to offset microvascular dysfunction and downstream reperfusion injury. Accordingly, IL-10 may primarily serve as a marker of injury severity and immune dysregulation rather than as a mechanistic contributor to futile recanalization.

Other biomarkers associated with futile recanalization in our study included IL-15 (pre-EVT timepoint) and SAA (post-EVT timepoint and intracranially in the sensitivity analysis). While human data on these markers remain limited, our findings are supported by preclinical evidence.^{29,30,47,48}

Regarding SAA, a study using transient middle cerebral artery occlusion in mice showed that, from its hepatic origin, SAA was rapidly induced after cerebral ischemia and contributed to ischemic brain injury by activating microglial inflammasomes, likely via the damaged BBB. At the same time, its genetic deletion led to significantly reduced infarct volumes.³⁰ In the human literature, a study focusing specifically on EVT-treated patients found that higher SAA levels, measured both immediately before

and 24 hours after the procedure, were independently associated with poor 3-month outcomes, reinforcing our findings.⁴⁹

Regarding IL-15, preclinical studies showed that it is induced in astrocytes, a finding also confirmed in brain tissue from patients with acute stroke.⁴⁸ Animal studies have further suggested that IL-15 appears to worsen ischemic injury by promoting the recruitment of CD8+T and natural killer (NK) cells.²⁹ In transgenic mice, astrocytic overexpression of IL-15 increased infarct size and neurological deficits, while its knockdown attenuated damage.^{47,48} Consistent with these results, our study is, to our knowledge, the first to report an association between elevated pre-EVT IL-15 levels and futile recanalization in humans. These findings also position IL-15 as a promising biomarker and therapeutic target in reperfusion injury.

In the present study, IL-6 was associated with futile recanalization at pre-EVT, intracranial, and 24-hour timepoints in univariate analysis, but these associations did not remain significant after adjustment for clinical confounders. This result contrasts with previous human studies.^{21,24,50} Mechtouff et al²⁴ reported an independent association between futile recanalization and IL-6 when sampled at 24 hours poststroke, though not at earlier timepoints. Similarly, in a study of the early MISO cohort, which included patients with incomplete recanalization or reocclusion, elevated intracranial IL-6 was associated with poor outcomes.²¹ Of note, the consistent univariate signal in our current study may suggest a loss of statistical power rather than a true absence of association. Mechanistically, IL-6 plays a role in reperfusion injury, as it has been shown to induce the expression of other biomarkers identified in our study, including VCAM-1 and SAA, and to modulate IL-10 responses.⁵⁰

Study Limitations

We acknowledge that (1) this was a single-center study. However, to our knowledge, this represents the largest prospective cohort to date combining peripheral and intracranial blood sampling during thrombectomy. (2) Given the exploratory, hypothesis-generating nature of our study, no formal sample-size calculation was performed. While the cohort was relatively large given the complexity of intracranial sampling, we caution that statistical power remained limited for detecting modest associations, especially in multivariable models. (3) Not all patients had samples available at every timepoint, due to evolving inclusion criteria over the course of the study (Figure S1), which may have further reduced statistical power, particularly at post-EVT and 24-hour timepoints. As an illustrative post hoc calculation, we could detect, in the smallest comparison set (24-hour sampling), a standardized mean difference of 1.15 (large effect size) with an 80% power and a 2-sided test at 0.05 significance level adjusted for the

37 biomarkers tested (after Bonferroni correction). (4) The observational design precludes causal inference regarding the relationship between biomarker levels and futile recanalization. For instance, higher IL-10 levels, despite its classically protective profile, may reflect a counterregulatory response to a more intense inflammatory burden and early immune dysregulation, rather than a direct harmful effect of IL-10. (5) Despite adjustment for key covariates associated with poor outcomes, residual confounding cannot be excluded. (6) Biomarker concentrations were assessed using a single multiplex immunoassay platform without orthogonal validation (eg, single-plex ELISA). While this approach provides internally consistent measurements across timepoints, multiplex assays may show limited cross-platform reproducibility for some analytes; therefore, independent confirmation of the key findings using alternative analytical methods is warranted. (7) Both baseline and follow-up imaging were heterogeneous (computed tomography versus MRI), which precluded robust, systematic infarct volume quantification across the full cohort; we, therefore, relied on ASPECTS as a pragmatic surrogate. (8) Markers of chronic cerebral small-vessel disease (white matter hyperintensities and cerebral microbleeds) were not systematically collected, precluding examination of the relationship between inflammatory biomarkers, cerebral small-vessel disease, and poor outcome.

Conclusions

Our study corroborates the clinical relevance of early assessment of blood-based biomarkers, VCAM-1, IL-10, IL-15, and SAA, for predicting futile recanalization. These immune-vascular factors, particularly when measured intracranially distal to the clot and peripherally immediately before or after EVT, could help identify patients at risk of poor outcomes despite successful recanalization and may also represent therapeutic entry points. Collectively, the processes outlined converge on endothelial activation, vascular inflammation, and BBB damage, supporting the hypothesis that microvascular dysfunction contributes to futile recanalization.

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Supplemental Material

Supplemental Methods
Tables S1–S3
Figures S1–S5
STROBE Checklist

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